Auditing a Released Back-Translation Evaluator for Sign Language Production: Replay Sensitivity and Public-Recipe (Non-)Reproducibility

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Abstract

Back-translation (BT) evaluators mediate sign language production rankings, but a released checkpoint may yield orderings that do not survive retraining or reference replacement. On the 641-sequence PHX-public subset of SLRTP2025, a constructed whole-donor replay probe (TN-PURE-v1) scores 23.02 sacreBLEU under the released BT evaluator versus 12.78 for the recorded test poses, a gap of **+10.24** BLEU points (95% CI $[\mathbf{+8.88, +11.62}]$). This constructed stress test is not reproduced by fourteen independently trained evaluators reconstructed from the public recipe: all fourteen show negative gaps (range $[\mathbf{-1.85, -0.80}]$; 95% prediction interval $[\mathbf{-1.94, -0.46}]$), and across 38 non-degenerate checkpoints from further training families (dev BLEU-4 4.5–10.9, all below the released evaluator's 13.38) every gap is strictly negative, decoded with a single canonical donor registry. On the matched 461-item confidence=1 subset of Czehmann et al.'s human back-translations, reference replacement attenuates the gap from **+10.03** to **+0.95** (90.5%), a reference-frame sensitivity. The released evaluator also decodes its 7,060-item training pool at 78.8 BLEU (70.7% exact match) while retaining dev/test performance (13.38/12.78); permutation, input-shuffle with output-equivariance verification, stratified exact-match, and same-signer held-out controls exclude a pose-ignorant ID/cache lookup. Because competence-matched replication was not possible, these results evidence public-pipeline non-replication for a constructed stress test — not a causal checkpoint effect, an identified memorization mechanism, or general leaderboard-rank instability.

Keywords: sign language production, back-translation evaluation, retrieval-augmented generation, multimedia quality assessment, evaluator non-replication

1 Introduction

Sign language production (SLP) systems are commonly evaluated by back-translation (BT): a trained sign recognizer decodes each output pose sequence, and a text metric compares the decoding with the reference. This measures recognizer-readable content through a learned measurement model, so system rankings may reflect both the evaluated outputs and the inductive biases of a particular evaluator checkpoint. The present audit is framed as a *public-pipeline reproducibility audit*: our primary research question is whether the released SLRTP2025 evaluator and its replay-probe reversal can be reproduced from the publicly documented training recipe. We do *not* frame this as a test of checkpoint identity, nor as a direct test of the Saunders et al. Saunders et al. (2021) rank-stability proposition: that proposition is conditioned on similar model configurations, whereas every checkpoint we can construct from the public recipe is less competent than the released evaluator (dev BLEU-4 4.5–10.9 vs. 13.38), so competence and checkpoint identity are confounded in our design and the non-reproduction result is scoped to the public recipe.

Prior work shows that BT scores can be modest for recorded real signing and can improve without faithful target conditioning Walsh et al. (2024); Hong and Košecká, Jana (2025), and sign-native or pose-aware metrics provide complementary evidence Kim et al. (2024); Imai (2025); Jiang et al. (2025). Nevertheless, BT remains central to benchmarks such as SLRTP2025 Walsh et al. (2025). An unresolved question is whether a retrieval–reference stress-test ordering persists across independently trained instances of the same BT architecture.

This question is exposed by a retrieval–reference reversal on PHX-public: under the original SLRTP2025 evaluator, source-conditioned whole-donor retrieval scores 23.02 sacreBLEU (0–100 scale), whereas the recorded test poses score 12.78. This neither establishes better signing by retrieval nor invalidates BT evaluation in general — it asks whether the reversal persists across evaluator checkpoints or depends on properties of the original evaluator.

We address this question and report three findings. *First, the reversal does not replicate across independently trained evaluators reconstructed from the public recipe*: fourteen same-architecture reconstructions yield a seed-level PURE–REC gap prediction interval of $[-1.94, -0.46]$ BLEU points, and 38 non-degenerate checkpoints from further training families (distillation, config-faithful, step-faithful, rescue) extend the negative range to $[-2.01, -0.21]$. *Second, the reversal is partly a property of the reference frame*: on the matched 461-item confidence=1 subset of public human sign-to-text back-translations Czehmann et al. (2025), the gap attenuates from +10.03 to +0.95 BLEU points (90.5%), an asymmetry that is inconsistent with simple floor-effect compression. *Third, the released evaluator exhibits a readout-with-generalization signature that no constructible checkpoint reproduces*: it decodes its 7,060-item training pool at 78.8 BLEU (70.7% exact-match) while retaining dev/test performance of 13.38/12.78, and a Gaussian weight-perturbation sweep preserves the reversal down to dev BLEU-4 ≈ 12.7—above the family ceiling of 10.9—before collapsing in lockstep with competence. The three findings are developed in $Results through four lines of evidence (conditioning controls, multi-checkpoint replications, reference replacement, and five diagnostic views), with the experimental stages serving as means to the findings rather than as a linear narrative. Evaluator-aware oracle analyses are relegated to SI Sup. H because their unmasked advantage does not survive reference-overlap exclusion (Fig. 1).

Why a constructed probe matters.

Retrieval-augmented SLP entries draw donors from the evaluator's training pool, so the TN-PURE-v1 stress test isolates a risk that production systems face in diluted form. Methodologically, the audit contributes a template — provenance table, checkpoint registry, evaluator-replication controls — that other benchmarks can adopt for public-recipe reproducibility testing.

2 Related Work

Learned evaluation in sign language production.

Neural SLP maps spoken language or glosses to pose and often evaluates with a frozen sign recognizer Camgoz et al. (2018); Saunders et al. (2020). BT was proposed as an SLP evaluation protocol by Saunders et al. Saunders et al. (2021), who argued the choice of BT model shifts absolute scores but not relative rankings between similar configurations; our audit does not directly test that proposition (its similarity condition is unmet in our reconstruction family) but characterizes a released evaluator on which a retrieval-probe reversal makes the proposition worth re-examining with matched evaluators. SignBLEU Kim et al. (2024), SiLVERScore Imai (2025), and pose-native measures offer complementary views, but BT BLEU remains central to benchmarks such as SLRTP2025 Walsh et al. (2025).

Retrieval and shortcut risks.

Retrieval-based SLP ranges from phonetic composition to retrieval-enhanced generation Walsh et al. (2024,?); Zuo et al. (2025); Tasyurek et al. (2025). The SLRTP2025 report describes a leading retrieval-based entry, making donor copying pertinent Walsh et al. (2025). More generally, memorization in retrieval-augmented language modeling shows that strong task scores can reflect reproduction of training examples Carlini et al. (2021); Kandpal et al. (2022); Khandelwal et al. (2020). A shortcut observed under one learned evaluator does not show that the resulting ordering persists under another checkpoint.

Robustness of learned metrics and corpus audits.

Hong and Košecká Hong and Košecká, Jana (2025) show BT-BLEU can diverge from target faithfulness; Walsh et al. Walsh et al. (2024) document a low ground-truth BT baseline; Jiang et al. Jiang et al. (2025) recommend BT likelihood over decoded BLEU. Yazdani et al. Yazdani et al. (2025), BackTranslation2.0 Cory et al. (2025), and He et al. He et al. (2024) expose further fragilities of model-based generation metrics. Two 2026 PHOENIX-2014T audits bear directly on our reversal: Czehmann et al. Czehmann et al. (2025) released human back-translations and documented reference information loss; Alkain et al. Alkain et al. (2026) showed PHOENIX-2014T has

Table 1: Three-way data provenance for PHX-public. The released evaluator's training manifest matches the released pose materialization (7,060/515/641), not the full official PHOENIX-2014T split. All 41 missing IDs share the reason `absent_from_slrtp2025_released_pose_records` (SI Sup. A). SHA-256 hashes verify file integrity.

Split source	Train	Dev	Test
Official PHOENIX-2014T	7,096	519	642
Released SLRTP2025 pose (`.pt`)	7,060	515	641
Released evaluator `config.yaml` manifest	7,060	515	641
Missing from official	36	4	1

the highest train–test text overlap among tested SLT corpora. Artiaga et al. Artiaga et al. (2025) showed signer-dependent evaluation overestimates SLT capability. None of these studies runs multi-checkpoint evaluator experiments or donor-pool substitution controls.

3 Methods

3.1 Evaluation design

Let E_c denote a BT evaluator checkpoint and $o_{i,n}$ the pose sequence for output condition i and item n. Checkpoint c produces $h_{c,i,n} = E_c(o_{i,n})$ and $B_{c,i} = M(\{(h_{c,i,n}, y_n)\}_{n=1}^{N})$, where y_n is the reference text and M is the corpus-level sacreBLEU functional unless stated otherwise. The values $B_{c,i}$ induce a ranking over fixed output conditions. We call a result *evaluator-pipeline-sensitive* when a substantive pairwise ordering changes across independently trained checkpoints under a fixed output set. The fixed primary set comprises recorded test poses (REC), random donor copy, source-conditioned whole-donor retrieval, a Progressive Transformer generator, and retrieval–generator compositions; oracle-gloss variants are diagnostic controls.

The main evaluation uses PHX-public, a German weather-domain text-to-pose benchmark. The released SLRTP2025 pose records contain 7,060/515/641 sequences; one official test ID is absent from the released materialization, so every result is conditioned on the complete 641-record release (SI Sup. A). Table 1 gives the three-way provenance: the official PHOENIX-2014T split, the released SLRTP2025 pose materialization, and the released evaluator's actual training manifest (confirmed via `config.yaml` which references the same `train.pickle/dev.pickle/test.pickle` that ship as `train.pt/dev.pt/test.pt`). Supplementary analyses use **PHX-holdout-MSKA** (133-joint, 1,538 windows) and CSL-Daily (Mode B, 1,176 sequences; SI Sup. L). Table 2 summarizes the protocols.

All pose sequences are evaluated at 12.5 fps, matching the SLRTP2025 evaluator training distribution; a three-path regression (canonical, explicit 25→12.5-fps, and an erroneous double-subsample path) confirms that no double subsampling occurs, and that the official Codabench starter-bundle Progressive Transformer reproduces the official 1.5942 BLEU under our pipeline (full implementation details in SI Sup. F).

Table 2: Experimental protocol summary.

Setting	Recognizer	Layout	Split size	N eval	α
PHX-public	SLRTP2025 evaluator	178-joint	641 seq	641	0.88 (tuned)
PHX-holdout-MSKA	mska_slr (phoenix-2014t)	133-joint	1,538 win	1,538	0.50 (frozen)
CSL-Daily	mska (csl-daily)	133-joint	256 of 1,176 win	256	0.50 (frozen)

PHX-public uses α=0.88 (selected on dev); PHX-holdout-MSKA and CSL-Daily use α=0.5.

3.2 Systems and retrieval controls

For each target, source-conditioned whole-donor retrieval scans the complete training pool after Unicode NFKC normalization, whitespace normalization and lowercasing, and selects the sequence with maximum source-text token-set Jaccard similarity. Oracle-gloss retrieval instead uses target gloss and is treated as an input-leakage diagnostic; random retrieval selects a donor uniformly. Donor exclusion removes exact normalized text or gloss duplicates or donors above threshold τ. Gloss-based exclusion uses target gloss and is therefore an offline diagnostic rather than a deployable inference procedure.

Text-nearest tie-breaking.

Jaccard ties are resolved by minimum character-level Levenshtein distance and then by minimum SHA-256 hash of the training key. The §3.2 algorithm also excludes donors whose NFKC-normalized text equals the query's (exact-normalized-text exclusion).

Single canonical materialization. Every number in this paper is generated from one canonical donor registry built by the deterministic §3.2 algorithm (exact-normalized-text exclusion applied), whose SHA-256 is `9170a53026ab3263b451a3632a9e02-318745acabf12f0b679921477afbefa301`. The artifact regenerates this registry and all headline numbers with a single command, writes them to a machine-readable `paper_numbers.json`, and checks the manuscript against it. An earlier pre-exclusion materialization (PURE 23.79, gap +11.01) survives only as a sensitivity analysis in SI Sup. N: on 610/641 items (95.2%) both materializations produce identical output, and a 14-reconstruction comparison gives mean $|\Delta\text{gap}| = 0.65$ BLEU (max 1.45), so qualitative conclusions are unchanged.

Retrieval composition.

TN-PURE-v1 uses whole-donor copy with [::2] subsampling to 12.5 fps, preserving the donor's original duration. TN-PTCOMP-v1 (a secondary control) composes a PT scaffold with donor hands and face via α-blending of upper-body joints (α=0.88 for PHX-public; full joint-set definitions and duration/word-count matching statistics in the artifact). The separately matched seen–unseen factorial retains 605/641 targets matched on duration and word count.

Donor-origin contrast retrieval systems.

To test whether the retrieval advantage co-occurs with donor origin, we construct frozen probe systems from the same 641 queries: **UNSEEN-PURE-v1** retrieves from the 640 other test poses (excluding the query itself, its sequence root, and exact-text matches); **CROSSFIT-PURE-A/B-v1** split queries by hash into 331/310 folds, each retrieving from the other fold; **SEEN-PURE-MATCHED-v1** selects a train-pool donor within ± 0.1 Jaccard of each UNSEEN donor (622/641 admit one); **SEEN-RAND640-MATCHED-v1** applies matched selection inside frozen 640-donor sub-pools. SEEN-PURE-v1 re-uses the canonical TN-PURE-v1 registry. Full construction rules (calipers, tie-breaks, signer constraints) are in the artifact; these are constructed probes, not deployable generators.

3.3 Evaluator checkpoint family

The primary inferential checkpoint family comprises the original frozen SLRTP2025 evaluator and six *recipe-conditioned reconstruction* runs trained from scratch with the documented architecture and a common frozen train/development partition (primary seeds 101–606; eight extension seeds 707–1405 use the identical protocol). The original bundle's `validations.txt` logs 202 validation points over 2,828 optimizer steps ($\approx$102 epochs), with decoded-BLEU selection reaching best dev BLEU-4 13.38 at step 1,820; intermediate checkpoints are not published. We use two reconstruction protocols: *paper-derived* (14 runs, legacy script, 300 epochs, NLL selection) and *released-config* (4 runs, up to 3,000 epochs, decoded-BLEU early stopping). Both share the documented architecture (3L/256/512ff, 8 heads), tokenizer, manifests (7,060/515), [:: 2] subsampling, Adam (1e-3, betas .9/.998, wd .001), batch size 256, and plateau $\times$0.8 LR schedule (full provenance in SI). A total of 70 training runs were completed across 13 families (full registry in SI Sup. D). Of these, 43 were decoded on PHX-public under the canonical §3.2 donor registry; five of these produce degenerate output (three α=1.0 distillation students with 641/641 or 483/641 empty hypotheses, and two smallest ladder fractions with BLEU $\approx$ 0) and are excluded from the non-replication count, which we therefore report over 38 non-degenerate checkpoints.

Readout-with-generalization signature (post-hoc operational definition).

A checkpoint exhibits the training-pool readout signature if it has both (i) training-pool free-decode BLEU $\geq$ 50 *or* exact-match $\geq$ 50%, and (ii) dev BLEU-4 $\geq$ 8.0. These thresholds were defined after observing the released evaluator (78.8 BLEU / 70.7% EM, dev 13.38) vs. the reconstruction family (max 11.5 train / 9.9 dev, no simultaneous satisfaction). The thresholds are operational heuristics for this case study, not validated criteria.

3.4 Metrics and uncertainty

The recorded test poses (REC) — the released recordings of human signing, which BT decodes and compares to text references — are decoded by each BT recognizer; we avoid the abbreviation "ground truth" because the recorded poses are inputs to a learned measurement model, not a semantic gold standard. We report corpus BLEU-4 on the 0–100 sacreBLEU scale as the primary metric Post (2018); sentence-level statistics (pass-through, slot-F1) are reported separately on the 0–1 scale. BLEU-1, chrF, ROUGE-L, BERTScore Zhang et al. (2020) (`bert-base-multilingual-cased`, no baseline rescaling, F1), and WER (jiwer 3.1.0 normalized) provide complementary decoded-text controls; BT likelihood and pose-DTWp provide non-decoded controls. A non-standard BLEURT substitute used only for a secondary reference-sensitivity check is documented, with its validation against canonical BLEURT-20 and its known bias, in SI Sup. O.

Scorer equivalence.

The vendored official scorer (SacreBLEU 1.4.2) and the paper scorer (SacreBLEU 2.5.1, `tok:13a`, `smooth:exp`) give identical corpus BLEU on all 28 evaluation cells (maximum $|\Delta| = 0.0000$; byte-identical hypotheses, all n-gram orders non-zero). We report the 2.5.1/13a signature as primary.

We report two forms of uncertainty. *Item-sampling uncertainty* uses paired sequence-level bootstrap (10,000 resamples, seed 42, percentile 95% CIs): per item, n-gram sufficient statistics are precomputed once, and every resample re-aggregates them into a corpus BLEU. Because queries reuse pose donors (565 unique donors for 641 queries), we complement the item bootstrap with donor-cluster and query$\times$donor two-way cluster bootstraps (SI Sup. G). A pairwise difference is called detectable when its interval excludes zero. Significance testing of the primary estimand uses a paired approximate-randomization permutation; because the headline gap was first observed in exploratory analysis and the estimand was frozen afterwards, we report a descriptive Monte-Carlo estimate: 0 of 10,000 permutations exceed the observed gap ($p \leq 10^{-4}$, exact 95% binomial interval $[0, 0.00037]$). *Training-replicate uncertainty* is reported only for the recipe-conditioned population: a descriptive seed-level 95% Student-t prediction interval (not extrapolable to the released evaluator). The eight extension seeds are post-hoc additions and enter only that interval, never the six primary comparisons. The single primary estimand is the PURE–REC decoded-sacreBLEU gap under the released evaluator; all other contrasts are secondary and are Holm-corrected within a defined family or explicitly labelled descriptive (SI Sup. C).

4 Results

4.1 Retrieval reverses the reference ranking under the original evaluator

Under the original SLRTP2025 evaluator, source-conditioned whole-donor retrieval scored 23.02 sacreBLEU (0–100 scale), compared with 12.78 for the recorded test poses — a retrieval–recorded-poses gap of +10.24 BLEU points (95% CI $[+8.88, +11.62]$; 10,000 sequence-level resamples). This fixed-checkpoint reversal motivates the analysis but does not indicate superior sign production.

Controls localized the observation to source-conditioned donor retrieval (Table 4): pure donor copy exceeded the oracle-gloss composed variant (11.8), canonical PT-composed (18.86), PT baseline (1.59), and 20-seed random-donor baselines (0.90

Table 3: Decoded-text metric sensitivity under the original SLRTP2025 evaluator (641 sequences). The reversal direction is consistent across decoded-text metrics; teacher-forced NLL and pose-DTWp do not reproduce it (full protocol details in SI Sup. F).

System	BLEU-4	BLEU-1	chrF	ROUGE-L	WER	BERTSc.
REC-v1 (recorded test poses)	12.78	34.40	34.59	35.20	85.77	0.757
PT-v1	1.59	16.91	19.03	17.26	101.68	0.669
TN-PTCOMP	18.86	46.35	41.88	45.17	77.36	0.793
TN-PURE	**23.02**	**55.43**	**47.82**	**53.07**	**71.87**	**0.822**
Gap PURE−REC	+10.24	+20.94	+12.93	+17.87	−13.90	+0.066

PURE, 0.77 COMP). Oracle-gloss retrieval (which uses reference gloss and is leakage) did not exceed the recorded test poses (11.4–11.8 vs 12.78). Token-normalized teacher-forced NLL (lower is better) does not reproduce the reversal: REC 3.060 vs PURE 3.479, COMP 3.411, PT 4.074. Likewise the normalized pose-DTW diagnostic does not validate semantics (REC 0 by construction vs PT/PURE/random 0.00678–0.00830). The reversal is visible under complementary text metrics on the same hypotheses (Table 3): sacreBLEU-4 +10.24, BLEU-1 +20.94, chrF +12.93, ROUGE-L +17.87, WER −13.90 (better), BERTScore-F1 +0.066. We describe the observation as a *decoded-text-metric reversal* and use sacreBLEU-4 as the primary decoded metric.

Metric implementations.

All secondary metrics use the official SLRTP2025 implementations: they reproduce the official repository values (BLEU-4 12.777, chrF 34.585, WER 85.775, ROUGE 35.196 Walsh et al. (2025)) exactly. Implementation choices matter for chrF (corpus-level sacrebleu `CHRF` defaults; a sentence-level variant yields 48.05 instead of 34.59) and WER (jiwer 3.1.0 with official normalization; raw whitespace-split yields 79.26 instead of 85.78). BLEU-4 is invariant; we fix the official implementations throughout (ablations in SI).

Decoding-parameter sensitivity.

Re-decoding across beam size $\{1, 3, 5\}$, length penalty $\{-1, 0\}$, and max length $\{30, 50\}$ shows the reversal is not an artifact of decoding: the PURE–REC gap is +9.9 (greedy), +10.2 (beam 3), +10.8 (beam 5), with length penalty and max length negligible.

Pure donor copy outperformed the composed variant for every source-conditioned retriever, and source-text Jaccard produced the highest pure-donor score (23.02, ahead of LaBSE 19.2 and multi-SBERT 16.5), while oracle-gloss retrieval did not exceed the recorded test poses in either construction (11.4–11.8 vs 12.78): the reversal does not require oracle-gloss input leakage, and pure copy — not composition — is its strongest form.

Table 4: PHX-public conditioning and construction controls (641 sequences). Columns distinguish whole-donor copy from PT-scaffold composition; all rows share the query set, donor pool, evaluator, and sacreBLEU protocol. Semantic retrievers (LaBSE, multi-SBERT) lie between the canonical retriever and oracle gloss (artifact).

Conditioning	Pure donor copy	PT+retrieval composed
Source text (German Jaccard)	**23.02**	18.86
Oracle gloss (input leakage)	11.4	11.8
Random donor (seeds 0–19)	0.90 (SD 0.24)	0.77 (SD 0.19)

4.2 Independent evaluator reconstructions do not reproduce the reversal magnitude

TN-PURE-v1 yields an original-evaluator gap of +10.24 BLEU points (95% CI $[+8.88, +11.62]$). Across fourteen reconstruction runs, all fourteen have negative PURE–REC gaps (range $[-1.85, -0.80]$). TN-PTCOMP-v1 yields an original gap of +6.09 $[+4.76, +7.41]$; all fourteen reconstruction estimates are negative. The seed-level 95% prediction interval is $[-1.94, -0.46]$ (mean −1.20, SD 0.33, 0/14 positive). For the six primary runs, original-minus-run difference-in-differences estimates range from +10.89 to +11.99; every 95% interval excludes zero.

4.2.1 Recipe-conditioned evaluator reconstructions

All fourteen reconstructions share frozen manifests, architecture, hyperparameters, and selection rule; seeds 101–1405 vary initialization and data order (Table 5). All evaluators are decoded with the canonical §3.2 donor registry (exact-normalized-text exclusion applied).

Table 5: Canonical fixed-output evaluation on the released 641-pose PHX-public subset, decoded with the §3.2 donor registry. PURE is TN-PURE-v1; COMP is TN-PTCOMP-v1. Gaps are relative to local REC (BLEU points). Per-cell values in the artifact.

Evaluator	REC	PURE	Gap	COMP	Gap
Original	12.78	23.02	**+10.24**	18.86	+6.09
Seed 101	9.24	8.45	−0.80	7.92	−0.65
Seed 202	8.22	6.92	−1.30	7.54	−1.28
Seed 303	9.10	7.25	−1.85	8.71	−0.96
Seed 404	9.52	7.71	−1.81	6.38	−0.81
Seed 505	9.71	8.44	−1.28	7.97	−1.08
Seed 606	7.55	6.31	−1.24	7.93	−0.16
Extension 707–1405 (range)	8.05–9.87	6.86–8.97	−1.48 to −0.87	6.90–8.83	−1.03 to −0.44

Competence is assessed on the released 515-sequence development subset, never on PHX-public for checkpoint selection. Original obtains dev BLEU-4 13.38 and WER 83.37 (official jiwer implementation, 0–100 scale); under the uniform full-pool beam-3 free-decode protocol, the 14 reconstruction runs obtain dev BLEU-4 6.5–9.4 and WER 88.21–94.83. Their selected validation NLL is 2.641–2.712 versus Original teacher-forced dev NLL 3.235, showing that likelihood and decoded recognition competence are not interchangeable. On PHX-public, reconstruction REC scores span 7.55–9.87 BLEU points, likewise below Original 12.78. Consequently, evaluator competence and unavailable original-training details remain confounded with evaluator identity; the experiment identifies evaluator-pipeline sensitivity, not a causal checkpoint-identity effect.

Competence gate and trajectory analysis.

A checkpoint is competence-matched only if both $|\Delta\text{dev BLEU-4}| \leq 1.0$ and $|\Delta\text{dev WER}| \leq 3.0$ (the released evaluator's dev values are 13.38 / 83.37). None of the 52 non-distillation trained checkpoints passes either half (dev BLEU-4 falls short by 3.4–8.9; dev WER exceeds the 86.37 bound by 0.08–13.3). The rescue search (batch size, dropout, weight decay, label smoothing, longer budget; Table 6) also fails: the best variant (weight-decay 0, seed 202) yields REC 9.88 / PURE 8.29 (gap -1.59). The gate conclusion is insensitive to threshold: across a $\pm[0.5, 2.0]$ BLEU $\times$ $\pm[1.5, 4.5]$ WER grid, 0 of 52 passes.

What the released training log establishes.

The bundle's `validations.txt` logs dev BLEU-4 reaching 13.38 at step 1,820 (Fig. 3a), while all constructible reconstructions plateau at 8–10 by epoch 50–100. The training dynamic that produced this slope is not explained by public artifacts. Input degradation (temporal downsampling) compresses both scores but does not eliminate the reversal (at k=8: REC 8.75, PURE 11.69, gap +2.94; SI Sup. F).

Table 6: Competence rescue sweep: 20 runs across six hyperparameter families all fail the competence gate. Best variant: weight-decay 0, seed 202 (gate threshold in §4.2.1).

Variant family	dev BLEU-4 range	min $\lvert\Delta\text{BLEU-4}\rvert$ vs Original
lr 5×10^{-4} / 2×10^{-3} (8 runs)	7.9–9.7	3.7
batch size 128 / 512	6.8–9.2	4.2
dropout 0.2	6.9–7.5	5.9
weight decay 0	**9.8–9.9**	**3.5**
label smoothing 0.1	6.8–7.4	6.0
long schedule (600 epochs, lr 5×10^{-4})	8.2–8.3	5.1

The six primary seeds carry precomputed paired-bootstrap intervals; the eight extension seeds reproduce the same pattern (seed-level 95% PI $[-1.94, -0.46]$).

Table 7: PURE–REC gap across metrics and evaluators on PHX-public (641 sequences; canonical decoding) for the original evaluator and the *six primary* reconstructions (of fourteen; the eight extension seeds show the same near-zero/negative pattern in Table 5). chrF, WER and ROUGE-L use the official SLRTP2025 implementations (§4.1). WER and NLL are sign-flipped so positive always means PURE is better than REC. BERTScore-F1 uses `bert-base-multilingual-cased` (bert-score 0.3.13 package, whose module `__version__` string reads 0.3.12; `lang=de`, `idf=False`, no baseline rescaling; model revision logged in the artifact); per-checkpoint BERTScore cells are in the artifact. The large reversal is consistent in direction across decoded-text metrics under the original evaluator and disappears under all six primary reconstructions; residual reconstruction signs are mixed and near zero for n-gram metrics.

Evaluator	ΔBLEU-4	ΔBLEU-1	ΔchrF	ΔWER†	ΔNLL†	ΔBERTSc.
Original	+10.24	+20.94	+12.93	+13.90	−0.419	+0.066
Seed 101	−0.80	−1.23	−1.44	−2.95	−0.219	−0.005
Seed 202	−1.30	−1.33	−1.57	−3.52	−0.233	−0.006
Seed 303	−1.85	−1.99	−2.26	−1.53	−0.240	+0.001
Seed 404	−1.81	−1.60	−1.90	−1.87	−0.194	−0.001
Seed 505	−1.28	−1.53	−1.59	−3.41	−0.268	−0.004
Seed 606	−1.24	−1.26	−1.55	−3.46	−0.169	−0.001

†WER and NLL are sign-flipped: positive means PURE is better than REC (lower WER, lower NLL).

Cross-metric consistency across checkpoints.

Table 7 reports the gap on the same fixed output under each of the seven primary evaluators across metrics. Under the original, the gap is positive and large for all decoded-text metrics; teacher-forced NLL is more negative for PURE than REC (−0.419 sign-flipped), consistent with the likelihood-vs-decoded-metric divergence Jiang et al. (2025). Under the six reconstructions the reversal disappears: all chrF and ROUGE-L gaps are negative, all WER gaps favor REC, and sacreBLEU-4 gaps are all negative (range $[-1.85, -0.80]$). The reversal is a property of decoded-text metrics under the original evaluator, not specific to sacreBLEU-4.

Rank stability (see SI Sup. H).

On a fixed nine-system stress-test set, the released evaluator ranks TN-PURE first; five of six reconstructions rank REC first, with moderate cross-checkpoint agreement (mean τ_b=0.44, ρ=0.57).

Reference-frame sensitivity: matched-subset analysis.

Two 2026 corpus audits bear directly on this reversal. Czehmann et al. Czehmann et al. (2025) released manual sign-to-text back-translations of the test set (one deaf fluent L2 DGS signer; per-item confidence flags), documenting information loss, lexical errors, and translationese in the original speech-transcribed references, and showed that reference replacement already changes BLEU scores and rankings on PHOENIX-2014T. Alkain et al. Alkain et al. (2026) quantified train–test text overlap across sign-language translation corpora, showing that a small fraction of highly training-like test items disproportionately inflates corpus BLEU. We reproduce the overlap pattern locally: the 32 test items whose reference exactly matches a training text score REC 40.26 / PURE 78.73 (gap +38.5), while the remaining 609 items score REC 11.93 / PURE 22.20 (gap +10.3).

Because Czehmann et al. compared original-reference and human-reference conditions on the same confidence=1 subset (462 of 642 items, excluding low-confidence, partial, or untranslatable items), we follow the same protocol. On the matched 461-item subset (after join with our 641 released test IDs), the original-reference gap is +10.03 BLEU points (REC 14.94 / PURE 24.98), and the human-reference gap is +0.95 (REC 7.35 / PURE 8.30), a paired attenuation of 9.08 BLEU points (90.5%, bootstrap CI $[7.44, 10.86]$; Table 8; Fig. 4; per-item paired view in SI Fig. S1). The residual human-reference gap is not distinguishable from zero (paired item bootstrap CI $[-0.13, 1.99]$, 10,000 resamples), so under the independent reference frame the reversal is not merely attenuated but statistically undetectable at the item-sampling level. Teacher-forced BT likelihood does not reproduce the reversal under either reference condition (REC NLL 3.060 vs PURE 3.479 on original refs; REC NLL 3.41 vs PURE 3.71 on human refs), consistent with the likelihood-vs-decoded-metric divergence Jiang et al. (2025). Scoring against both references jointly (multi-reference, closest-length convention) leaves the gap at +11.50 — *larger* than the original-reference gap. This is expected under the retrieval criterion: TN-PURE donors are selected by lexical overlap with the source sentence, and the original reference is a speech transcription of that same sentence, whereas the human back-translation is an independent sign-conditioned paraphrase. Adding the original reference to the reference set therefore injects a second lexically coupled target that the replay decode can match, while the recorded poses' decode matches neither reference well; the multi-reference gap inherits the lexical coupling rather than averaging it away. The reversal is thus specific to reference frames containing the original speech-transcribed register. The full-641 comparison (+10.24 $\rightarrow$ +0.86, attenuation 91.6%) is reported as sensitivity analysis; it mixes reference replacement with floor effects from low-confidence items and is not the primary estimand. This is a *descriptive contrast of two corpus gaps*, not a causal attribution: the attenuation cannot distinguish reference-text defects, paraphrase/translationese shift, BLEU floor effects, disruption of the lexical coupling between retrieval criterion and original reference, or evaluator training-distribution alignment. Under all six reconstruction checkpoints the gap against human references is negative (-0.15 to -0.65). The asymmetry is consistent across metrics on the matched subset: chrF attenuation 82.2% (+12.73 $\rightarrow$ +2.27), BLEU-1 attenuation 84.0% (+20.01 $\rightarrow$ +3.19). Czehmann et al. also showed BLEURT is less sensitive to reference replacement than BLEU; our BLEURT-substitute analysis agrees and is reported with full validation details in SI Sup. O. We emphasize what this analysis does not show: it quantifies how the decoded text scores shift when the reference frame is changed, not whether the replay donor motion itself conveys target-equivalent meaning. The human back-translations are sign-conditioned paraphrases of the recorded test poses, not judgments of the replay donors, so the attenuation bears on the lexical scoring frame and leaves the semantic adequacy of TN-PURE-v1 unassessed.

A natural concern is that the attenuation is a symmetric floor effect. On the matched subset, switching from original to human references, REC drops 7.59 BLEU-4 points (50.8%) but PURE drops 16.68 points (66.8%) — PURE loses a larger fraction (ratio 1.31×). The asymmetry is even larger under BLEU-1 and chrF, so simple symmetric compression is insufficient, but the asymmetry does not discriminate between reference-text defects and training-distribution alignment.

Table 8: Matched-subset reference sensitivity (corpus sacreBLEU, BLEU points) under the released evaluator. Primary analysis: confidence=1 subset (461 items) with original and human references computed on the *same* items. Sensitivity analysis: full 641 items. Human references: Czehmann et al. Czehmann et al. (2025) back-translations (CC BY-NC-SA 4.0). Confidence breakdown: 461 confidence=1.0, 121 confidence=0.5, 59 confidence=0.0.

Reference condition	*N*	REC	PURE	Gap
Primary: matched confidence=1 subset				
Original (speech-transcribed)	461	14.94	24.98	+10.03
Human back-translation (conf=1)	461	7.35	8.30	+0.95
Paired attenuation		9.08 BLEU pts (90.5%), CI [7.44, 10.86]		
Sensitivity: full 641 items				
Original (speech-transcribed)	641	12.78	23.02	+10.24 [+8.88, +11.62]
Human back-translation (full)	641	5.66	6.52	+0.86 [+0.38, +1.91]
Paired attenuation				9.38 BLEU pts (91.6%)
Human refs, reconstructions (6 seeds)	641	3.49–4.62	3.35–4.35	−0.15 to −0.65

Table 9: Asymmetric floor-effect control (full-641 sensitivity analysis). PURE loses a larger fraction of its score
REC across all metrics when references are replaced, so simple symmetric compression is insufficient; the asymm
does not discriminate between reference-text defects and training-distribution alignment.

Metric	REC (orig)	REC (human)	REC drop	PURE (orig)	PURE (human)	PURE d
BLEU-4	12.78	5.66	7.11 (55.7%)	23.02	6.52	16.50 (71.
BLEU-1	34.40	29.28	5.12 (14.9%)	55.43	32.35	23.08 (41.
chrF	34.59	32.75	1.84 (5.3%)	47.82	35.02	12.80 (26.
Dual-ref gap (orig+human)	BLEU-4: +11.50; BLEU-1: +20.03					

Table 10: Donor-origin contrast estimands (sacreBLEU, BLEU points; gaps vs local REC). SEEN = 7,060 training poses; UNSEEN = 640 other test poses. Per-seed cells in the artifact.

Evaluator	SEEN-PURE	UNSEEN-PURE	SEEN−REC	UNSEEN−REC	Estimand
Original	23.02	8.51	+10.24	-4.27	**+14.51**
Seeds (range)	6.83–9.09	6.59–7.98	−1.85 to −0.80	−1.94 to −0.27	−0.15 to +1.22

4.3 Donor-origin contrast characterizes the reversal's co-occurrence with training-pool donors

If the released-evaluator reversal requires exposure to training-pool donors, then restricting retrieval to poses the released evaluator has not seen should remove or invert the advantage. Under the original SLRTP2025 evaluator, UNSEEN-PURE-v1 (retrieving from the 640 other test poses) scores 8.51 sacreBLEU versus 12.78 for REC, a gap of −4.27 BLEU points: the +10.24 advantage of SEEN-PURE-v1 does not survive retrieval-pool substitution. The donor-pool substitution estimand (SEEN − REC) − (UNSEEN − REC) = +14.51 BLEU points is large but is a *donor-pool substitution effect*, not a pure exposure effect, because SEEN-PURE-v1 and UNSEEN-PURE-v1 differ simultaneously in pool size (7,060 vs 640), median Jaccard (0.50 vs 0.37), donor reuse (Gini 0.11 vs 0.29), and training-pool exposure. Cross-fit controls reproduce the same direction (CROSSFIT pooled −3.8 BLEU points vs fold-local REC). Under the six reconstructions all pool-origin contrasts lie in $[-0.15, +1.22]$ (Table 10).

Path-dependent decomposition and balanced factorial.

We decompose the substitution estimand descriptively along telescoping paths (SI Sup. E). The path-based origin estimate varies from +5.8 to +8.4 BLEU across admissible orderings because size and selection interact. The better-balanced 605-query factorial (matching seen/unseen donors at high/low LaBSE similarity within predeclared tolerances) estimates the pool-origin main effect at +2.08 $[+1.56, +2.57]$, the similarity main effect at +6.96 $[+5.93, +8.08]$, and the seen×similarity interaction at +3.70 $[+2.63, +4.79]$ (all Holm p=0.0003). Across reconstructions, the seen main effect shrinks to +0.51 and the interaction is null (−0.39). Two conclusions: Jaccard-only paths inflate the pool-origin component, and the released evaluator's seen advantage *grows with donor–query similarity* while reconstruction interactions are null. The caveat: seen donors remain more similar than unseen within strata (SMD +0.4), so this is a train-pool vs test-pool provenance contrast, not an identified training-exposure effect.

4.4 Competitive-system context from the public leaderboard

Because the audit's stress-test set is constructed rather than competitive, the public SLRTP2025 leaderboard offers useful external context. The leaderboard was computed under the same released evaluator we audit, and its "Ground Truth" row reproduces our canonical REC numbers exactly (BLEU-4 12.78, BLEU-1 34.40, WER 85.77; Walsh et al. Walsh et al. (2025), Table 1). Table 11 reproduces the ph14t-test scores. The retrieval-based winner, USTC-MoE, sits roughly even with REC on the lexical BT metrics — 0.72 BLEU-4 below, 0.45 BLEU-1 and 2.24 chrF above, with worse WER (93.49 vs 85.77) — while its DTW-MJE (0.0448) and duration ratio (1.438) indicate considerable pose-level deviation from recorded signing. The co-occurrence of approximately matched lexical BT scores with markedly poorer pose fidelity admits several readings: that lexical BT under this evaluator is insensitive to pose fidelity, that template-dense retrieval is well matched to PHX-public's register, or that DTW-MJE itself undersells the production. The observation does not, by itself, establish that the released evaluator favors retrieval entries; resolving that question would require re-decoding fixed competition outputs under multiple evaluators. We note one naming distinction: the leaderboard's "Progressive Transformer" row (BLEU-4 5.43) is the original Saunders et al. Saunders et al. (2020) model scored by the challenge organizers, whereas our PT-v1 control (BLEU-4 1.59) is the official Codabench starter-bundle checkpoint shipped with the SLRTP2025 release; the two are different checkpoints of the same architecture, and the gap between them illustrates the sensitivity of BT-BLEU to training and selection details even within a single model family.

Table 11: Public SLRTP2025 challenge leaderboard on ph14t-test (Walsh et al. Walsh et al. (2025), Table 1), computed under the same released BT evaluator we audit; the "Ground Truth" row matches our canonical REC exactly. USTC-MoE is roughly even with REC on lexical BT metrics (BLEU-4 −0.72, BLEU-1 +0.45, chrF +2.24, WER +7.72) despite a much larger DTW-MJE (0.0448) and duration ratio (1.438). This single-evaluator co-occurrence is descriptive and does not establish evaluator retrieval-favoring bias.

System	BLEU-1	BLEU-4	chrF	ROUGE	WER↓	DTW-MJE↓
Recorded test poses (REC)	34.40	12.78	34.59	35.20	85.77	0.0000
Team 1 – USTC-MoE (retrieval)	**34.85**	**12.06**	**36.83**	**36.59**	93.49	0.0448
Team 3 – Hacettepe	30.44	9.59	29.70	30.64	88.88	0.0423
Progressive Transformer	22.17	5.43	24.13	21.98	101.45	0.0418
Team 2 – hfut-lmc	16.96	2.05	25.88	19.77	147.85	0.0403
TN-PURE-v1 (our replay probe)	55.43	23.02	47.82	53.07	71.87	0.00678

Scope limit and cross-evaluator re-decoding.

The public challenge artifacts include only the Progressive Transformer baseline output (`PT_baseline_test.pt`, which we evaluate as PT-v1); the per-team pose outputs are not archived, so the competition systems cannot be re-decoded under our reconstruction family. Whether the roughly even lexical-BT scores of USTC-MoE and REC survive across independently trained evaluators is therefore an open empirical question that this audit cannot resolve. We note, as a benchmark-organizer responsibility, that archiving fixed per-team outputs alongside public scores would make rank-stability audits feasible for third parties.

4.5 Local perturbation of the released evaluator at competence parity

The reviewer's competence confound (released dev BLEU-4 13.38 vs. reconstructions ≤ 10.9) cannot be broken by retraining. We can, however, perturb the released checkpoint *locally*: starting from the released weights, we add i.i.d. Gaussian noise to every parameter (excluding BatchNorm running statistics, which must remain positive) at scale $\sigma \in \{0, 10^{-3}, 3\times10^{-3}, 10^{-2}, 3\times10^{-2}\}$ with two seeds each, and re-decode PHX-public REC/PURE and the dev set under the canonical donor registry (Fig. 6). Small σ perturbs checkpoint identity while preserving competence, so this design operates where the 14-seed reconstruction family cannot — at and below the released competence from the released checkpoint itself.

The reversal is preserved under competence-preserving perturbation and collapses in lockstep with competence. At $\sigma \leq 3\times10^{-3}$ the dev BLEU-4 stays at 12.76–13.35 (above every reconstruction's maximum of 10.9) and the PURE–REC gap stays at +9.50 to +10.35 (within ±0.7 of the baseline +10.24). The collapse happens only at $\sigma \geq 10^{-2}$, where dev BLEU-4 simultaneously degrades to 7.3–10.2 (gap +3.2 to +4.2) and at $\sigma = 3\times10^{-2}$ both dev BLEU-4 (≈ 0.1) and the gap (≈ 0) are extinguished together. Two conclusions follow. First, the reversal is not a fragile identity-level artifact: at competence parity it is locally robust. Second, the reversal tracks competence rather than checkpoint identity: it survives perturbations that preserve competence and disappears only when the noise destroys competence. This shifts the unobserved interval: the reconstruction family places a lower bound at dev BLEU-4 10.9 (no reversal), and the perturbation sweep shows preservation down to dev BLEU-4 ≈ 12.7, narrowing the unobserved transition to roughly $[10.9, 12.7]$ rather than $[10.9, 13.4]$.

Epoch-checkpoint decoupling search (experiment D).

We additionally searched the four available intermediate epoch checkpoints (`rescue/seed_202_wd0` and `long_schedule/seed_202` at epochs 25 and 50, plus their best checkpoints) for a decoupling point that would break the within-family readout–gap coupling — e.g., elevated train-pool readout but PURE–REC gap still ≤ 0. No such point exists: all six checkpoints stay in the (low readout, negative gap) quadrant, with train-pool BLEU 5.5–8.9 (1.6–2.0% EM) and PURE–REC gap -0.94 to -1.70 (per-checkpoint values in `results/epoch_decouple.json`). The within-family relationship between low readout and negative gap is therefore monotone across the entire observable trajectory.

4.6 Per-item gap decomposition: what kind of item drives the reversal

To characterize the *mechanism* rather than the identity question, we regress the per-item sentence-level PURE–REC gap y_i (sacreBLEU sentence BLEU-4 under the released evaluator, canonical donor registry; mean 8.4, SD 18.7, $n = 641$) on interpretable item-level features: donor–reference token-set Jaccard, normalized character Levenshtein between donor and reference, an exact-match-train-text flag, the number of training texts sharing the reference's 60-character prefix (template density), the number of training texts with Jaccard ≥ 0.5 to the reference (high-overlap-train), reference text length, donor reuse count, a confidence=1 flag, and a 9-level signer one-hot block (Fig. 7). The full model explains $R^2 = 0.33$ of the per-item variance. The strongest single-feature contributions, measured by leave-one-out R^2 drop, are high-overlap-train (0.093), template density (0.051), and donor–reference Levenshtein (0.039); the donor–reference Jaccard contributes 0.007, the exact-match flag 0.001, and the entire signer block 0.008 — signer identity is essentially irrelevant.

The signs of the standardized coefficients are informative. Items where the donor is lexically close to the reference (high Levenshtein similarity, $\beta^* = +6.1$; high donor–reference Jaccard, $\beta^* = +2.9$) show *larger* gaps, whereas items with many high-overlap training neighbors ($\beta^* = -7.6$) or high template density ($\beta^* = -7.3$) show *smaller* gaps. The reversal is therefore largest on items where the retrieval criterion (source-text Jaccard) happens to deliver a donor that matches the reference register closely, but where the recorded poses do not independently let the recognizer recover that text — precisely the lexical-coupling mechanism predicted by the construction (the original reference is a speech transcription of the same source sentence the donor was selected by). Signer identity and text length contribute negligibly. The unexplained 67% of variance is consistent with per-item pose variability and BT-decoder stochasticity that these surface features do not capture. This decomposition does not identify a hidden-pipeline detail; it localizes the reversal to a lexical-coupling property that is constructible on PHX-public's template-dense, high train–test overlap register, consistent with the corpus-level audit of Alkain et al. Alkain et al. (2026).

4.7 What distinguishes the released evaluator: five converging diagnostic views

The first finding (negative gaps across reconstructions) raises a positive question: what property of the released evaluator do the reconstructions lack? We approach this through five diagnostic views that triangulate on a single signature rather than independently confirming it (Fig. 3b,c; Table 12; full version in SI Sup. D). The views reuse the same released checkpoint, PHX-public data, and related donor constructions; we therefore report them as descriptive triangulation, not as five independent tests.

Readout* with *generalization: the distinctive combination.

The released evaluator decodes its full 7,060-item training pool at free-decode BLEU 78.8 with 70.7% exact-match — strong readout — while retaining non-trivial dev/test performance (13.38/12.78). Under the identical full-pool, beam-3 free-decode protocol, the constructible family stays far below: 14 reconstruction seeds span train BLEU 7.2–9.3 (EM 1.5–2.3%) at dev BLEU 6.5–9.4, and 12 non-degenerate distillation students span train BLEU 6.6–8.9 (EM 1.3–1.9%) at dev BLEU 7.0–9.0 (these free-decode dev BLEU-4 values are not comparable to the training-time dev BLEU-4 reported in Table 13, which was recorded under the original training protocol — checkpoints were

Table 12: Summary of the five diagnostic views. The released evaluator sits far outside the family range on all five views. Per-view family composition in SI Sup. D.

Diagnostic view	Family n	Released evaluator	Family range	Gap to family max
(i) Gap–competence	43	+10.24 at dev 13.38	$[-2.01, 0.00]$ at dev 4.5–10.9	+10.24
(ii) Pass-through ratio	24	3.09 (sentence)	1.05–1.31	+1.78
(iii) Train-pool readout BLEU	27	78.8 (70.7% EM)	6.6–9.3 (1–2% EM)	+69.5
(iv) Frame-reversal loss	1	−16.1 (PURE)	−4.3 (wd0 rescue)	|11.8|
(v) Competence gate	52	13.38	4.5–10.02 (max)	+3.36

selected by dev NLL, SI Sup. E — and reaches 10.9); an overfit rescue endpoint (epoch checkpoint) reaches high prefix-level training-decode BLEU but collapses on dev NLL (SI Sup. F). None of the 27 non-degenerate constructible checkpoints approaches the released evaluator's 78.8/70.7%. Its donor-content pass-through is a 2.4× outlier (ratio 3.09 vs. family 1.05–1.31).

Leakage sanity checks for the train-pool readout.

The 78.8/70.7% readout is not a trivial implementation artifact. Five controls (full data in SI Sup. M) rule out four classes of shortcut: teacher forcing (code-path audit confirms autoregressive beam search), output–reference misalignment (label permutation collapses BLEU from 78.6 to ∼1.0), pose-ignorant ID/cache lookup (input-side pose permutation with verified output equivariance collapses BLEU to <1, and a same-signer held-out control reaches only BLEU 12.19 / EM 3.2% on unseen poses versus 78.8 / 70.7% on training poses), and narrow-text memorization (high exact-match across all 9 signers and on unique training texts). The checks do not identify the training mechanism but are consistent with pose-conditioned training-pool exposure. The thresholds defining the signature were set after observing the released evaluator and are operational heuristics, not validated criteria (§3.3).

Knowledge-distillation competence ladder.

To probe the unobserved competence interval above the reconstruction ceiling (dev BLEU-4 10.0), we trained 15 distillation students from the released evaluator as teacher: five distillation strengths ($\alpha \in \{0, 0.25, 0.5, 0.75, 1.0\}$) crossed with three seeds, with loss $\mathcal{L} = (1-\alpha)\,\mathrm{CE} + \alpha\, T^2\, \mathrm{KL}(\text{teacher}\|\text{student})$ at T=2. All 15 completed training and beam-3 PHX-public evaluation. Table 13 shows the distilled students extend the observable range upward (dev BLEU-4 up to 10.9), yet all twelve α<1.0 students have negative PURE–REC gaps (−2.01 to −0.77). The three α=1.0 students produce empty or degenerate outputs (seed 101: 641/641 empty; seed 202: 483/641 empty, 158 single-token "."; seed 303: 641/641 empty), yielding near-zero REC and PURE BLEU and an approximately zero gap. Training-pool readout (uniform full-7,060 protocol) stays at 6.6–8.9 BLEU / 1.3–1.9% EM across the twelve α<1.0 students versus the released evaluator's 78.8/70.7%. Full soft-label distillation (α=1.0) degenerates into empty or near-empty outputs and does not constitute a valid transfer test. Under the tested distillation procedure (soft-label KL at T=2 across the α grid), the training-pool readout signature therefore does not transfer.

Donor-registry consistency note.

All checkpoints in this paper are decoded with the single canonical §3.2 donor registry. The earlier pre-exclusion materialization (PURE 23.79, gap +11.01) and its per-item provenance analysis (30 of 31 differing items explained by the exclusion rule) are reported only as a sensitivity analysis in SI Sup. N; a 14-reconstruction sensitivity comparison shows mean $|\Delta\text{gap}|$ = 0.65 BLEU (max 1.45).

Table 13: Per-α distillation ladder summary. 15/15 students evaluated with the canonical donor registry from §3.2; "dev BLEU-4 (max)" is the training-time dev BLEU-4 recorded by the original training protocol (legacy validation-selected protocol; not comparable to the uniform full-pool free-decode dev BLEU-4 reported in §4.7; checkpoints were selected by training-time dev NLL, SI Sup. E). α=0 = hard-label only; α=1 = full soft-label KL. All α<1.0 yield negative gaps; α=1.0 produces empty/degenerate outputs (641/641 or 483/641 empty hypotheses). Train-pool readout uses the identical full-7,060, beam-3, [:: 2]-subsample protocol for every student and for the released evaluator (no 1,000-item-prefix mixing); the three α=1.0 students decode to empty or near-empty strings on both dev and test (uniform full-pool: 515/515 or 390/515 dev hypotheses empty, 641/641 or 483/641 test hypotheses empty) and score 0.00/0.0%; their dev BLEU-4 is degenerate and not reported.

α	n	dev BLEU-4 (max)	gap (range)	train readout (BLEU / EM%)	reaches 10.5?
0.0 (hard-label)	3	10.51	$[-1.71, -0.80]$	8.45–8.87 / 1.9%	1/3
0.25	3	10.75	$[-2.01, -0.77]$	7.81–8.63 / 1.5–1.9%	2/3
0.5	3	10.41	$[-1.38, -0.77]$	7.73–8.20 / 1.7–1.9%	0/3
0.75	3	10.93	$[-1.21, -0.81]$	6.59–7.01 / 1.3–1.4%	2/3
1.0 (full KL)	3	*degenerate*	$[0.00, 0.00]$	0.00 / 0.0% (empty)	—
Released (full 7,060, beam=3)	1	13.38	+10.24	78.82 / 70.7%	—

Competence extrapolation (exploratory).

Three fits (OLS linear, OLS quadratic, GP with RBF + white kernel) over the 43 checkpoints spanning dev BLEU-4 $[4.5, 10.9]$ all predict negative gaps at 13.38, with substantial model sensitivity (OLS quadratic flips from −13.5 to +1.3 when distillation is removed; GP upper bound stays below +2.0 across leave-one-family-out folds). We treat this as an exploratory visual guide (full fit table in SI Sup. K), not as evidence that competence alone cannot explain the reversal. The unobserved interval is narrowed to roughly $[10.9, 12.7]$ by the released-checkpoint perturbation sweep (§4.5), which shows the reversal preserved down to dev BLEU-4 $\approx$ 12.7; non-linear behavior inside this interval cannot be excluded.

Synthesis.
Across the five diagnostic views, the released evaluator is the only checkpoint combining high training-pool readout (78.8 BLEU, 70.7% EM) with retained in-domain performance (13.38/12.78 dev/test). No reconstruction or distilled student reproduces either the +10.24 gap or this training-pool readout signature.

4.8 Checkpoint-targeted oracle stress tests (SI-diagnosed)

We treat evaluator-aware donor selection as a checkpoint-targeted oracle stress test, not as a deployable SLP method, and we report its unmasked advantage as a non-robust diagnostic: the diagonal advantage does not survive reference-overlap exclusion (SI Sup. H). The full cross-checkpoint transfer matrix, candidate-pool sweep, and selector/evaluator-label permutation test are likewise in SI Sup. H because they share target references and evaluator by design. The main text states only that the unmasked advantage is not robust to reference-overlap exclusion and that the white-box search budget (candidate-pool size) controls whether any decoded-BLEU margin is detectable.

5 Discussion

Three-sentence summary.
A retrieval–reference reversal of +10.24 BLEU points under a released BT evaluator does not survive scrutiny: it disappears under independent reconstructions of the same evaluator family, attenuates by 90.5% under independent human references, and co-occurs with a training-pool readout signature that is unique to the released checkpoint within our constructible family. The non-replication is scoped to the public training recipe because all constructible checkpoints remain below the released evaluator's competence, leaving an unobserved interval that the released-checkpoint perturbation sweep narrows but does not eliminate. These three findings reframe the reversal as a property of one released pipeline rather than a property of BT evaluation in general, and they motivate a template for rank-stability auditing that other benchmarks can adopt.

The audit yields a broad negative result and a descriptive positive characterization. The negative result is carried by the fourteen same-recipe reconstruction seeds (shared architecture, tokenizer, manifest, selection rule; seeds 101–1405), whose Student-t prediction interval for the PURE–REC gap is $[-1.94, -0.46]$ BLEU points; beyond this replication family, 38 non-degenerate checkpoints from heterogeneous families (distillation, ladder, config-faithful, step-faithful, rescue, confirmation; all decoded with the canonical donor registry from §3.2) all have strictly negative gaps (range $[-2.01, -0.21]$) and are reported as exploratory dose–response data, not as IID replicates of a single training process (SI Sup. D). The positive characterization is that the released evaluator carries a high training-pool readout combined with retained in-domain performance that no public artifact in our family exhibits (§4.7). Because all constructible checkpoints have lower competence than the released evaluator (family maximum dev BLEU-4 10.9 vs. released 13.38), this non-reproduction is observed under a competence confound — but the released-checkpoint perturbation sweep (§ 4.5) shows the reversal is preserved down to dev BLEU-4 ≈ 12.7 at competence parity, narrowing the unobserved transition to roughly $[10.9, 12.7]$ and ruling out a fragile identity-level artifact. An exploratory OLS/GP extrapolation is reported in SI Sup. K for visualization only, without inferential claims. Because Saunders et al. Saunders et al. (2021) recommended BT for comparing between similar model configurations, and our evaluator family is additionally competence-unmatched (all checkpoints dev BLEU-4 < 10.9 vs. released 13.38), our non-reproduction result is scoped to the public training recipe and does not constitute a strong test of their rank-stability proposition.

On the matched 461-item confidence=1 subset, human-reference substitution descriptively attenuates the gap by 90.5% (+10.03 $\rightarrow$ +0.95, paired bootstrap CI for attenuation $[7.44, 10.86]$); the asymmetric PURE-vs-REC drop ratio (1.31$\times$ fractional for BLEU-4 on the matched-461 subset; 1.29$\times$ on the full-641 sensitivity set) is inconsistent with a simple symmetric compression model but does not discriminate between reference-text defects and training-distribution alignment as non-exclusive candidate mechanisms. Czehmann et al. Czehmann et al. (2025) already showed that reference replacement changes BLEU scores and rankings on PHOENIX-2014T; our contribution on this axis is applying the matched-subset protocol to the released SLRTP2025 evaluator and quantifying paired attenuation, not establishing reference-set sensitivity as a novel phenomenon. Because TN-PURE-v1 reuses pose donors across queries (641 queries select 565 unique donors; 61 donors serve more than one query, maximum reuse 6; Gini 0.11), item-level resampling overstates the effective sample size; we therefore additionally resample at the donor level. The headline gap is stable under all cluster-aware resamplings (SI Sup. G): donor-cluster (565 clusters, CI $[8.87, 11.64]$), query$\times$donor two-way ($[8.79, 11.74]$), signer (9 clusters, $[9.29, 11.39]$; percentile coverage is unreliable at this cluster count, so this is a sensitivity reference only), broadcast-date (297 clusters, $[8.83, 11.72]$), and signer$\times$show (17 clusters, $[7.89, 11.91]$) are all consistent with the query-level CI $[8.88, 11.62]$; the broadcast-show split has only 2 clusters and its interval is reported in SI Sup. G as an illustrative sensitivity check without confirmatory weight.

Relation to prior work.

These findings extend prior critiques of BT validity from output faithfulness to public-recipe reproducibility: improved BT scores need not imply target conditioning Hong and Košecká, Jana (2025), low recorded-pose BT baselines are documented Walsh et al. (2024), and BT likelihood is a more stable correlate of human judgment than decoded BLEU Jiang et al. (2025). Our reversal is a *decoded-sacreBLEU* reversal; teacher-forced BT likelihood does not reproduce it (REC NLL 3.060 vs PURE 3.479), consistent with that divergence. Whole-donor replay is a constructed probe, whereas SOKE-style word-level retrieval can improve pose quality under a different representation Zuo et al. (2025); and multiple text scorers on the same BT decoding are not independent confirmation.

Validation frequency is an underappreciated hyperparameter.

Step-faithful vs epoch-validation shows validation frequency alone shifts the final checkpoint by 1.5–2.0 BLEU points (SI Sup. F).

Independent reproduction with greedy decoding.

Under greedy decode the released evaluator scores REC 12.24 and PURE 22.12 (gap +9.88); no trained checkpoint reproduces the reversal. Qualitative conclusions are identical to beam-3 (artifact at `results/cells_greedy/`).

What would be needed to identify the mechanism.

Identifying the mechanism requires the SLRTP2025 organizers to disclose (i) the exact training command and seeds, (ii) the data pipeline beyond [::2] subsampling, (iii) any curriculum or checkpoint averaging, and (iv) intermediate checkpoints. A leave-one-donor-out retrain sweep is computationally tractable (∼1,400 GPU-hours) but requires the original training infrastructure, which is not publicly available.

Actionable recommendations for benchmark organizers.

The audit yields three transferable lessons, scoped to the constructed replay-probe stress case on PHX-public rather than to general SLP leaderboard stability.

Lesson 1: a released evaluator is a research artifact, not an infrastructure component. The released `config.yaml` alone was insufficient for 70 same-recipe runs to reach competence parity, and the readout-with-generalization signature could not be localized to a specific training stage because intermediate checkpoints are unpublished. Benchmarks should release (i) the exact training command and seeds, (ii) data pipeline scripts beyond the manifest, (iii) a SHA-256 manifest of training tensors, (iv) intermediate checkpoints at least every 100 epochs so that trajectory-level analyses can localize when distinctive properties emerge, and (v) the fixed per-team pose outputs alongside the public scores. The current release provides only the Progressive Transformer baseline output; without archived per-team outputs, rank-stability audits across independently trained evaluators—like the one in this paper—cannot be reproduced by third parties on real competition systems.

Lesson 2: retrieval-based entries require shortcut-robustness certification. A donor-exclusion sweep is cheap, deterministic, and exposes the retrieval advantage. Leaderboard entries that retrieve from the training pool should attach $\tau \in \{0.3, 0.5\}$ donor-exclusion results as a shortcut-robustness certificate. Two descriptive probes further separate genuine signing from pipeline artifacts: the training-pool free-decode BLEU ÷ dev free-decode BLEU ratio (released: 6.1; reconstructions: 1.1–1.4) flags anomalous training-pool alignment, and the input-side pose permutation with fixed IDs and output-equivariance verification (SI Sup. M) excludes a pose-ignorant ID/cache lookup. The released evaluator passes the latter but fails the former by a wide margin.

Lesson 3: reference validity is part of evaluation validity. The 90.5% attenuation under Czehmann et al. Czehmann et al. (2025) back-translations shows that rankings computed against a single speech-transcribed reference can misrepresent sign-language quality. Benchmarks should adopt human back-translations as a secondary reference set, follow the matched-subset protocol (confidence=1 items) when comparing original-reference and human-reference conditions, and flag systems whose gap diverges sharply between the two. The cost is one additional reference file and a paired bootstrap; the benefit is detection of an entire class of reference-coupling artifacts that single-reference BLEU cannot surface.

6 Limitations

The audit is confined to PHX-public, a template-dense weather domain with documented reference-validity and train–test-overlap issues Czehmann et al. (2025); Alkain et al. (2026). The fixed nine-system stress-test set comprises replay, composition, weak generation, and random controls rather than competitive SLP systems; the public SLRTP2025 leaderboard (§4.4) offers only single-evaluator context — the retrieval-based winner reaches lexical-BT scores roughly even with REC despite far poorer pose fidelity — and the per-team pose outputs are not publicly archived, so those real systems cannot be re-decoded under the reconstruction family. No causal claim about evaluator retrieval-favoring follows from the leaderboard alone, and the audit characterizes a constructed stress case rather than a general leaderboard-rank instability across evaluators.

All 67 non-holdout checkpoints are less competent than the released evaluator (family maximum dev BLEU-4 10.9 vs. released 13.38); the original training history is unpublished, so competence-matched replication from the public recipe was not possible. The released-checkpoint perturbation sweep (§ 4.5) narrows the unobserved transition to roughly dev BLEU-4 $[10.9, 12.7]$, but non-linear behavior inside that interval cannot be excluded. Saunders et al. Saunders et al. (2021) limited their rank-stability claim to similar model configurations; our result does not directly test their claim under those conditions. The specific mechanism is not identifiable from public artifacts: neither recipe-conditioned reconstruction, rescue sweeps, config-faithful retraining, nor any tested distillation procedure (soft-label KL at T=2 across the α grid; the α=1.0 full-KL recipe degenerates into empty outputs and is not a valid transfer test) transfers the property. The donor-pool substitution decomposition is path-dependent; pool-origin contrasts are descriptive associations, not identified training-exposure effects. The human back-translations are a single-annotator derivative resource whose high-confidence subset lacks second-annotator adjudication.

The reversal is a decoded-BT-text-metric reversal, not a sign-language quality reversal. Without target-conditioned DGS signer judgment or a validated sign-native pose metric, we cannot determine whether the retrieved donor motion conveys target-equivalent meaning; we explicitly do not claim that donor motion fails to express the target, nor that the reversal indicates dataset contamination or pose memorization.

Use of public data.

All experiments use publicly released datasets (PHOENIX-2014T, CSL-Daily, SLRTP2025) and Czehmann et al.'s human back-translations. Pose tensors may retain signer identity; we release no per-signer raw poses.

7 Conclusion

This audit reports three findings on a +10.24 BLEU-point retrieval–recorded-poses reversal under a released BT evaluator on PHX-public. The reversal does not replicate across 38 non-degenerate checkpoints from independent training families (14 same-recipe reconstructions plus 24 heterogeneous-family checkpoints; gap range $[-2.01, -0.21]$, all decoded with a single canonical donor registry). It attenuates by 90.5% on a matched confidence=1 subset when references are replaced by independent human back-translations. And it co-occurs with a training-pool readout signature (78.8 BLEU / 70.7% EM with retained dev/test performance of 13.38/12.78) that no constructible checkpoint exhibits; input-side pose permutation with output-equivariance verification rules out a pose-ignorant ID/cache lookup.

The scope is correspondingly bounded. A Gaussian weight-perturbation sweep preserves the reversal down to dev BLEU-4 ≈ 12.7 and collapses in lockstep with competence, narrowing—but not eliminating—the unobserved interval between the family ceiling (10.9) and the released checkpoint (13.38). The training log shows validation PPL rising from 14.84 at step 546 to 25.42 at the best-BLEU step 1820, indicating that decoded-BLEU selection chose a checkpoint with higher validation NLL than the NLL minimum. Because competence-matched replication was not possible from public artifacts, the findings evidence public-pipeline non-replication for a constructed stress test, not a causal checkpoint effect, an identified memorization mechanism, or general leaderboard-rank instability.

8 Compliance with Ethical Standards

Conflict of interest.

The authors declare no conflict of interest.

Ethical approval.

All experiments use publicly released datasets (PHOENIX-2014T, CSL-Daily, SLRTP2025) and Czehmann et al.'s human back-translations. No animal research was conducted.

Funding.

[FUNDING-PLACEHOLDER: list grants or state "The authors declare that no funds, grants, or other support were received during the preparation of this manuscript."]

Data availability.

All numerical values, per-checkpoint data, decoded hypotheses, and analysis scripts are in the anonymous artifact at https://anonymous.4open.science/r/rcp-audit-artifact-B314/README.md; a persistent DOI-bearing release will be deposited upon acceptance. Upstream datasets and their licences: PHOENIX-2014T (CC BY-NC-SA 4.0, PHOENIX-2014T homepage); SLRTP2025 released pose records and the released BT checkpoint (challenge terms, SLRTP2025 repository); CSL-Daily Mode B (available for academic research under the CSL-Daily licence, CSL-Daily homepage); Czehmann et al. Czehmann et al. (2025) human back-translations (CC BY-NC-SA 4.0). Pose tensors may retain signer identity; we release no per-signer raw poses, only de-identified decoded text and aggregate metrics. The `bert-base-multilingual-cased` model is used under the Apache-2.0 licence of its release.

Supplementary Information (SI)

The following supplementary materials, referenced throughout the main text, are available as a separate file (`supplementary.pdf`) and in the anonymous artifact. Each appendix is annotated with the question it answers.

- Sup. A: Missing Test ID Sensitivity — could the single absent test item change the headline gap? (`sec:appendix_missing_id`)
- Sup. B: Per-seed Extension Cells 707–1405 — does the prediction interval hold beyond the six primary seeds?
- Sup. C: Analysis Provenance — what is the evidentiary role and multiplicity handling of each reported analysis? (`sec:provenance`)
- Sup. D: Canonical Checkpoint Registry — what exactly was trained, and which families enter which claim? (`tab:checkpoint_registry`)
- Sup. E: Donor-Pool Substitution Decomposition — how much of the seen-pool advantage is pool-origin versus donor similarity? (`sec:appendix_decomposition`, `tab:path_sensitivity`)
- Sup. F: Config-Faithful and Perturbation Details — does the reconstruction plateau reflect a protocol mismatch, and how robust is the reversal to weight perturbation? (`sec:appendix_config_faithful`, `sec:appendix_perturbation`, `tab:perturbation`)
- Sup. G: Cluster-Aware Bootstrap — does the gap survive resampling by signer, broadcast-show, and date? (`sec:appendix_cluster_bootstrap`)
- Sup. H: Supplementary Oracle Diagnostics — is the evaluator-aware advantage robust to reference-overlap exclusion? (`sec:appendix_oracle`)
- Sup. I: Template-Density, Holdout Boundary, Donor-Exclusion — does the reversal persist on a clean template-holdout split and under donor-text exclusion? (`sec:appendix_template_boundary`)
- Sup. J: Transcript-Slot Diagnostic Details — do slot-level semantic scores reproduce the BLEU reversal? (`sec:appendix_slot`)
- Sup. K: Competence Extrapolation — could a competence gap in the unobserved interval explain the reversal? (`sec:appendix_extrapolation`)
- Sup. L: CSL-Daily Boundary Control — does the readout signature generalize beyond PHX-public? (`sec:csl-daily`)
- Sup. M: Leakage Sanity Checks — can the train-pool readout be explained by implementation-level shortcuts? (`sec:appendix_leakage`)
- Sup. N: Pre-Exclusion Materialization Sensitivity — how sensitive are the headline numbers to the donor-exclusion rule? (`sec:appendix_materialization`)
- Sup. O: BLEURT-Substitute Validation — does a learned metric corroborate the BLEU finding under reference replacement? (`sec:appendix_bleurt`)

Checkpoint Registry (summary)

Table 14: Canonical checkpoint registry. 70 training runs across 13 families; 38 non-degenerate checkpoints enter the non-replication count, 43 have a canonical PHX-public PURE–REC gap, and the released evaluator is the audit target. Full per-checkpoint details in SI Sup. D.

Family	Planned	Trained	Has gap	Has dev	Enters
Reconstructions (14)	14	14	14	14	multiseed, bootstrap, PI, dose–resp.
Rescue (20)	20	20	1	20	gate; wd0 dose–resp.
Other families (21)	21	21	14	20	dose–resp.
Distillation (15)	15	15	15	15	ladder, dose–resp.
Total	70	70	43	67	
Released original	—	—	1	1	audit target

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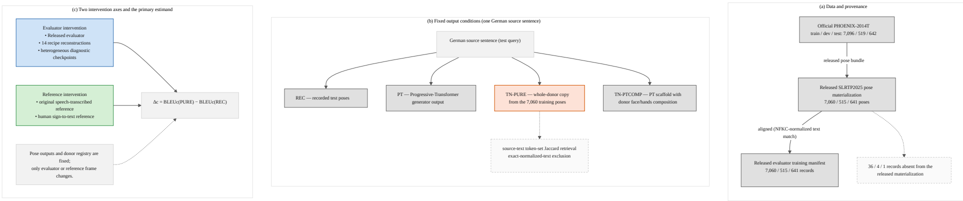

Fig. 1: Audit design: a fixed set of pose outputs is scored under two interventions. (a) Data and provenance: the official PHOENIX-2014T split (7,096/519/642) was materialized by the SLRTP2025 release as 7,060/515/641 poses; the released evaluator's training manifest aligns with the released pose materialization, with 36/4/1 official records absent from the release. (b) Fixed output conditions for one German source sentence: the recorded test poses (REC), the Progressive-Transformer generator output (PT), the whole-donor copy TN-PURE (donor retrieved by source-text token-set Jaccard, exact-normalized-text exclusion), and the PT-scaffold composition TN-PTCOMP are all evaluated identically. (c) Two intervention axes: the evaluator (released, fourteen recipe reconstructions, heterogeneous diagnostic checkpoints) and the reference frame (original speech-transcribed vs. human sign-to-text) are varied while pose outputs and the donor registry are fixed; the primary estimand is $\Delta_c = \text{BLEU}_c(\text{PURE}) - \text{BLEU}_c(\text{REC})$.

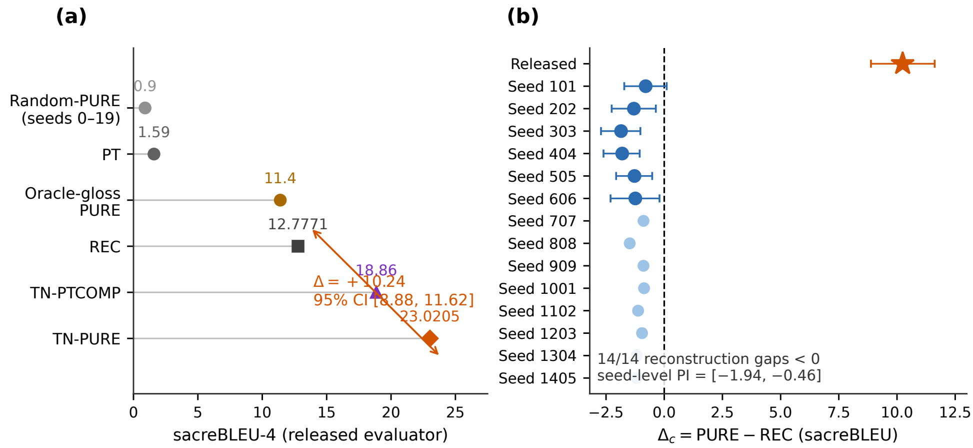

Fig. 2: Headline reversal under the released evaluator and its absence across recipe reconstructions. (a) System scores under the released evaluator (sacreBLEU-4, canonical 641-query protocol): TN-PURE 23.02 vs. REC 12.78 ($\Delta = +10.24$, 95% CI $[8.88, 11.62]$), with TN-PTCOMP 18.86, PT 1.59, oracle-gloss PURE 11.4, and random-donor PURE 0.90 (mean over seeds 0–19, ± 1 SD). (b) Per-checkpoint PURE−REC gaps: the released evaluator (+10.24) vs. fourteen recipe reconstructions, all negative (primary seeds with paired-item bootstrap 95% CIs; seed-level 95% PI $[-1.94, -0.46]$). All reconstructed evaluators are less competent than the released evaluator; this figure demonstrates public-pipeline non-replication, not an isolated causal checkpoint effect.

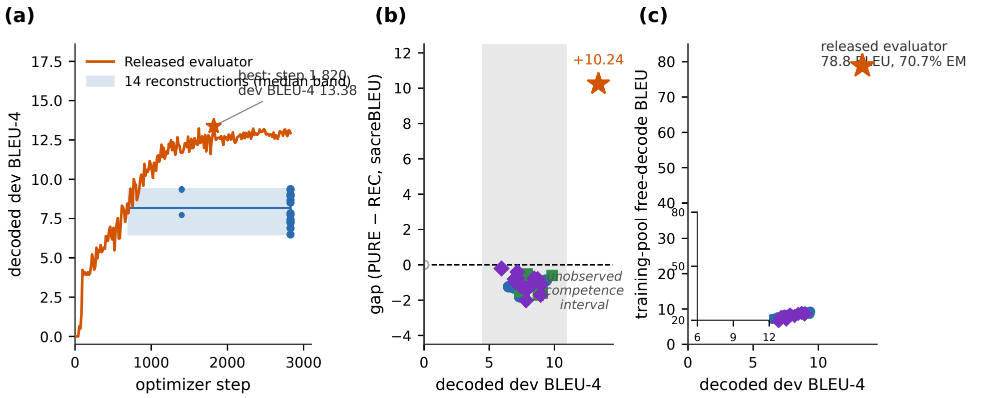

Fig. 3: Decoded competence and training-pool readout separate the released evaluator from every constructible checkpoint. (a) Dev-BLEU-4 training trajectories on a common optimizer-step axis: the released evaluator (orange) reaches 13.38 at step 1,820 of 2,828 (its archived validation log, 202 logged points); fourteen reconstructions form a median band (blue) with per-seed best points, plus uniformly decoded epoch-25/50 points (step = epoch × 28). (b) Canonical PURE−REC gap vs. decoded dev BLEU-4 by family: reconstructions (circles), config-faithful/step-faithful/confirmation/long-schedule (squares), rescue (triangle), distillation students (diamonds), released evaluator (star); the gray band marks the observed 4.5–10.9 range and the (10.9, 13.38) interval is unobserved; degenerate checkpoints (empty decodes) are hollow gray points. (c) Full-training-pool free-decode readout (beam-3, full 7,060 poses, uniform protocol) vs. decoded dev BLEU-4: every constructible checkpoint reads out 5.9–9.4 BLEU, while the released evaluator reads out 78.8 BLEU (70.7% exact match) at dev 13.38; the inset zooms the family region. Families are descriptive; checkpoints are not IID replicates.

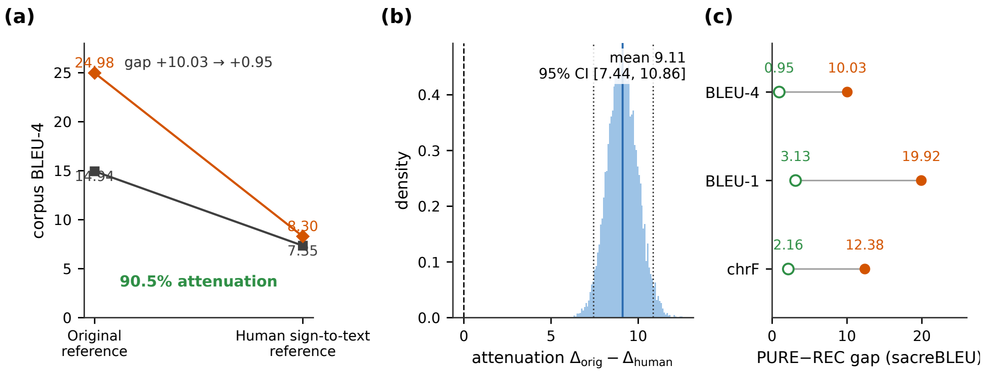

Fig. 4: Human sign-to-text references remove most of the replay advantage on the matched high-confidence subset ($n = 461$; released evaluator, canonical hypotheses). (a) Corpus BLEU-4 under original vs. human references: REC 14.94 $\rightarrow$ 7.35, PURE 24.98 $\rightarrow$ 8.30; the gap attenuates $+10.03 \rightarrow +0.95$ (90.5%). (b) Bootstrap distribution of the attenuation contrast (10,000 paired item resamples; mean 9.08, 95% CI $[7.44, 10.86]$; dashed zero line). (c) PURE−REC gap under original (solid) vs. human (hollow) references across BLEU-4, BLEU-1, and chrF. References are back-translations by a single deaf fluent L2 DGS signer Czehmann et al. (2025); reference replacement cannot distinguish reference-text defects, paraphrase/translationese shift, BLEU floor effects, or disruption of the lexical coupling between the retrieval criterion and the original reference.

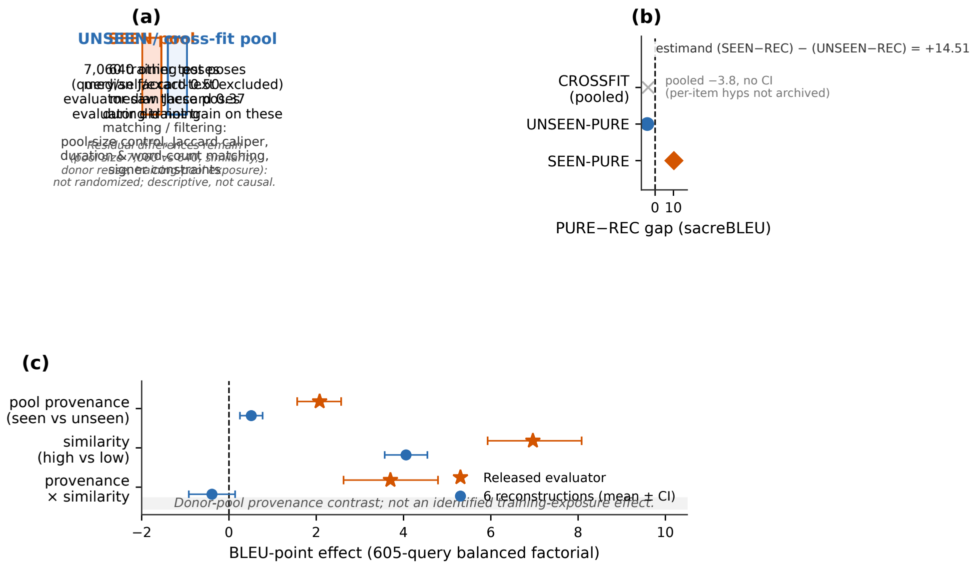

Fig. 5: The replay advantage co-occurs with training-pool donor provenance, but provenance and similarity remain confounded. (a) Pool construction: SEEN retrieves from the 7,060 released training poses (median donor–reference Jaccard 0.50; the released evaluator trained on these poses), UNSEEN from 640 other test poses (median Jaccard 0.37; the evaluator did not train on these), with pool-size control, Jaccard caliper, duration and word-count matching, and signer constraints; residual differences (pool size, similarity, donor reuse, training-pool exposure) remain, so the contrasts are descriptive, not causal. (b) Gaps relative to local REC: SEEN-PURE +10.24; UNSEEN-PURE −4.27 (paired-item bootstrap CI [−5.39, −3.20]); the pool-substitution estimand (SEEN − REC) − (UNSEEN − REC) = +14.51; CROSSFIT pooled −3.8 reported in the text without CI (per-item hypotheses not archived). (c) Balanced 605-query factorial (12.5-fps corrected protocol): released evaluator (star) vs. six-reconstruction means (circles) for the pool-provenance main effect, the similarity main effect, and their interaction. Donor-pool provenance contrast; not an identified training-exposure effect.

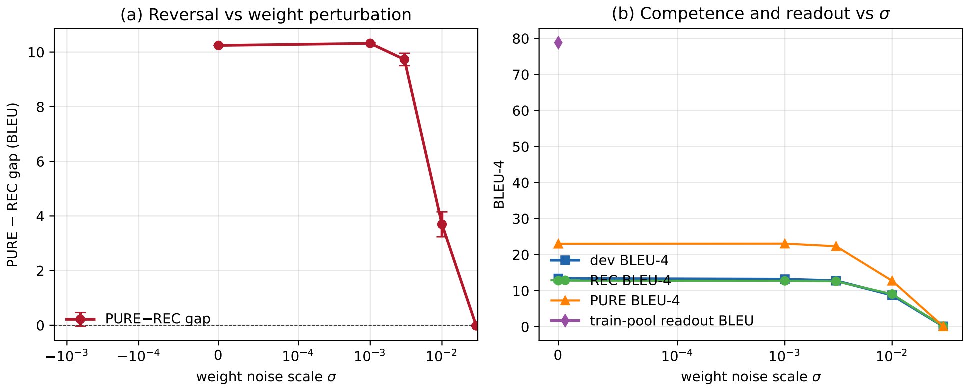

Fig. 6: Released-checkpoint weight perturbation (experiment A1). (a) PURE–REC gap vs. Gaussian weight-noise scale σ (two seeds per $\sigma > 0$; baseline at $\sigma = 0$). The reversal is preserved within ± 0.7 of $+10.24$ for $\sigma \leq 3\times 10^{-3}$, where dev BLEU-4 stays above 12.7 (above every reconstruction's maximum of 10.9). (b) Dev BLEU-4, REC BLEU-4, PURE BLEU-4, and (at baseline) train-pool readout BLEU-4. The reversal and competence collapse together at $\sigma \geq 10^{-2}$; $\sigma = 10^{-1}$ produces an all-empty-decoding model and is omitted. The reversal is locally robust at competence parity and tracks competence, not identity.

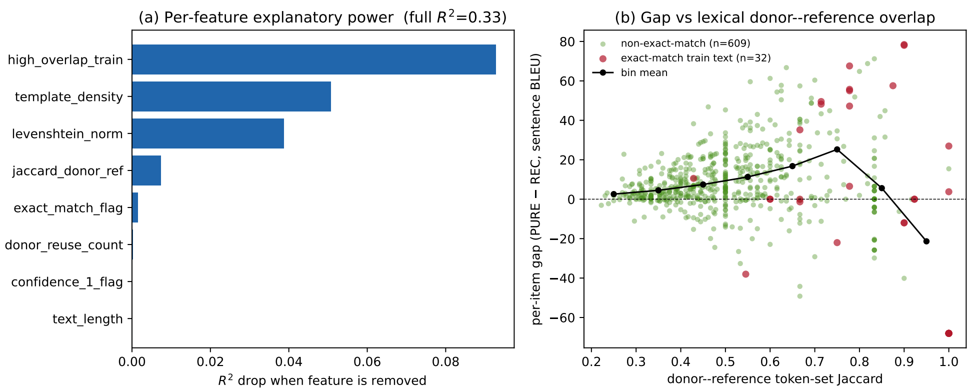

Fig. 7: Per-item gap decomposition (experiment C). (a) Per-feature R^2 drop when the feature is removed from the full OLS model ($R^2 = 0.33$, $n = 641$). High-overlap-train count and template density are the strongest single contributors; the full signer block contributes $\leq 1\%$. (b) Per-item gap vs donor–reference token-set Jaccard; red marks the 32 items whose reference text exactly matches a training text, green the remaining 609; the black line is the bin mean. The gap grows with donor–reference lexical overlap, consistent with the lexical-coupling mechanism. Outcome is sentence-level BLEU-4 (sacreBLEU 2.5.1, 13a, exp smooth); sentence-level mean need not equal corpus BLEU.